

DATA HANDOFF — CLEANED TABLES

Albion Care Network — Bed Demand Forecasting

From: Data Analysts (Team 5) · To: Data Science track · Re: modelling-ready inputs

Cover Note

The six source tables have been cleaned in Power Query and exported as CSVs for the modelling work. Files are in the shared clean-data folder:

/data/clean/

Each file mirrors exactly what feeds the Power BI operational dashboard, so dashboard figures and model inputs come from the same cleaned base. A full record of every cleaning decision and its rationale is in the accompanying Data Cleaning Documentation. Key points to be aware of before modelling:

- **Length of Stay:** in admissions, use the derived LOS_hours (recomputed from admission/discharge timestamps). The original length_of_stay_hours column was incomplete and has been dropped.
- **Structural nulls kept on purpose:** time_to_admission (ED) is null for non-admitted patients, and cancellation_reason (surgery) is labelled "Not Cancelled" for completed cases. These are correct — please do not impute them.
- **Filled gaps:** expected_los_days was filled with the per-procedure-group median (not a global value). occupied_beds nulls were forward-filled by hour.
- **De-duplication:** 486 fully-identical duplicate rows were removed from admissions (row count 131,479 → 130,993). 37 rows share an admission_id but conflict on other fields; these were retained and flagged — treat with care if joining on admission_id.
- **Negative LOS:** a small number of admission records had discharge before admission (invalid). See the cleaning doc for how these were handled; filter or exclude as appropriate for modelling.
- **Date range:** data spans 2024-01-01 to early 2026 (a few admission/discharge dates fall in Jan–Feb 2026). 5 hospitals only — this is a sample, not the full network.

Join key across all tables is hospital_id (links to hospital_reference). Suggested next step your side: align temporal granularity (hourly occupancy vs daily staffing) as noted in the plan.

Data Dictionary — Cleaned Tables

Grain and key columns for each cleaned file. “Cleaned?” flags where a column was changed from source.

bed_inventory_occupancy_clean.csv — grain: hospital × ward × bed_type × hour

Column	Type	Description	Cleaning applied
datetime	datetime	Hour of observation	Typed (UK locale)
hospital_id	text	FK → hospital_reference	—
ward	text	Ward name	—
bed_type	text	Critical Care / Standard / Day Case	—
total_beds	int	Physical beds	—
staffed_beds	int	Beds available to patients (KPI denominator)	—
occupied_beds	int	Beds in use	Nulls forward-filled by hour
closed_beds	int	Unavailable beds	—

admissions_discharges_clean.csv — grain: one row per admission

Column	Type	Description	Cleaning applied
hospital_id	text	FK → hospital_reference	—
patient_id	text	Patient identifier	—
admission_id	text	Admission identifier	486 exact dupes removed; 37 conflicts retained
admission_datetime	datetime	Admission time	Typed (UK locale)
discharge_datetime	datetime	Discharge time	Typed (UK locale)
admission_type	text	Emergency / Elective / Transfer	—
admission_source	text	Source of admission	—
specialty	text	Clinical specialty	—
ward	text	Ward	—
bed_type	text	Bed type	—
LOS_hours	decimal	Derived length of stay (hours)	NEW: discharge – admission
discharge_destination	text	Where patient went	Blanks labelled “Unknown”

ed_outpatient_arrivals_clean.csv — grain: one row per ED/outpatient arrival

Column	Type	Description	Cleaning applied
hospital_id	text	FK → hospital_reference	—
arrival_id	text	Arrival identifier	—
arrival_datetime	datetime	Arrival time	Typed (UK locale)
triage_category	text	Triage level	—
arrival_mode	text	How patient arrived	—
outcome	text	Admitted / Discharged / etc.	—
time_to_admission	decimal	Wait to admission	Null for non-admitted — keep null
bed_requested	text	Bed requested flag	—

elective_surgery_schedule_clean.csv — grain: one row per surgery

Column	Type	Description	Cleaning applied
hospital_id	text	FK → hospital_reference	—
surgery_id	text	Surgery identifier	—
surgery_date	date	Scheduled date	Typed
specialty	text	Specialty	—
procedure_group	text	Procedure category	—
expected_los_days	decimal	Expected stay (days)	Nulls filled: per-group median
bed_type_required	text	Required bed type	—
surgery_status	text	Completed / Cancelled	—
cancellation_reason	text	Reason if cancelled	Blanks labelled “Not Cancelled”

staffing_resource_clean.csv — grain: hospital × ward × role × day

Column	Type	Description	Cleaning applied
date	date	Staffing day	Typed (UK locale)
hospital_id	text	FK → hospital_reference	—
ward	text	Ward	—
staff_role	text	Role	—
planned_staff	int	Planned headcount	—
actual_staff	int	Actual headcount	—
safe_ratio_met	text	Yes / No	—

hospital_reference_clean.csv — grain: one row per hospital (dimension)

Column	Type	Description	Cleaning applied
hospital_id	text	Primary key	—
hospital_name	text	Hospital name	—
city	text	City	—
tier	text	flagship / large / medium	—
specialty_emphasis	text	Main specialty	—
total_bed_capacity	int	Total capacity	—

Questions on any cleaning decision → see the Data Cleaning Documentation, or message the DA. Please flag before re-cleaning any column so we keep one source of truth for the clean layer.